

MP1 — Small Language Model Challenge

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This report presents an empirical study on improving a small GPT-style language model trained from scratch on the WikiText-2 dataset. Starting from a provided baseline (width=128, heads=4, depth=4, approximately 2.10 test BPB), I systematically explored architectural variations. I found that increasing model width from 128 to 256 and extending training to 2400 steps reduced the validation BPB from 2.071 to 1.689 and test BPB to 1.7117, while increasing depth or attention head count did not help. An ablation study confirmed that the improvement is specifically attributable to the increased width. I also explored extending training from 1200 to 2400 steps, which further reduced BPB to 1.606, but this configuration was rejected because its evaluation time exceeded the 5× baseline scoring limit. The final submitted model (width=256, 2400 steps) achieves a good balance between prediction quality and computational cost, and is fully compliant with all resource constraints.

My Modifications

I created a custom configuration configs/my_config.json and systematically varied the following:

Depth: 4 → 8

Width: 128 → 256

Head count: 4 → 8 (at width=256)

Width (exploration): 256→512 (rejected)

Training steps (boundary exploration): 1200 → 2400 (at width=512)

All other hyperparameters (seed, context length, tokenizer, data) were kept identical to the baseline to ensure controlled comparisons.

To verify that the observed improvement is due to width rather than any other factor, I performed an ablation study: I reverted width to 128 while keeping all other changes (same student.py, same training procedure). If the score returns to the baseline level, this confirms width is the key mechanism.

Experimental Results

Experiment	width	heads	depth	Steps	Params	Val BPB	Test BPB	Eval Time
Baseline	128	4	4	1200	1.09M	2.071	2.101	~6 s
Deeper (D=8)	128	4	8	1200	1.88M	2.073	—	—
Wider (W=256)	256	4	4	1200	3.94M	1.863	1.893	~15 s
More Heads (H=8)	256	8	4	1200	3.75M	1.878	—	—
Wider (W=512)	512	4	4	1200	13.79M	1.682	1.7012	~63 s
Wider + Longer (W=512, 2400步)	512	4	4	2400	13.79M	1.606	1.6265	~78 s
Ablation (W=128)	128	4	4	1200	1.09M	2.071	—	~6 s
Final (W=256 + 2400 步)	256	4	4	2400	3.94M	1.689	1.7117	~23 s

Key observations:

1. Depth did not help: increasing depth from 4 to 8 slightly worsened validation BPB (2.071 $\rightarrow$ 2.073).
2. Width significantly helped: increasing width from 128 to 256 improved validation BPB to 1.863; further increasing to 512 improved it to 1.682 — an 18.8% improvement over the baseline.
3. Ablation confirmed width is the cause: reverting width to 128 (with everything else unchanged) brought BPB back to exactly 2.071, matching the baseline.
4. More heads did not help: at width=256, increasing heads from 4 to 8 slightly worsened BPB (1.863 $\rightarrow$ 1.878).
5. Boundary exploration: extending training to 2,400 steps at width=512 reduced validation BPB to 1.606, but its evaluation time ($\sim$ 78 s) exceeded the 5 $\times$ baseline scoring limit ($\sim$ 30 s). Therefore, this configuration was not selected. The final submission instead uses width=256 + 2,400 steps, which achieves comparable BPB (test 1.7117) while satisfying all resource constraints.

The width=512 + 2400 steps model achieved the lowest BPB (1.606), but its evaluation time ($\sim$ 63 s) exceeded the 5 $\times$ baseline scoring limit ($\sim$ 30 s). This is a deliberate trade-off analysis:

Configuration	Val BPB	Test BPB	Eval Time
W=256 + 2400 steps (final)	1.689	1.7117	$\sim$ 23 s
W=512 + 1200 steps	1.682	1.7012	$\sim$ 63 s
W=512 + 2400 steps	1.606	1.6265	$\sim$ 78 s

The assignment explicitly requires models to respect the evaluation budget. I therefore rejected the lower-BPB configuration in favour of the compliant one, even though it costs $\sim$ 4.5% higher BPB. This reflects a principled trade-off between prediction quality and computational cost.

Conclusion

I improved the baseline model's test BPB from 2.101 to 1.7117 (an 18.5% reduction) by increasing model width from 128 to 256 and extending training from 1200 to 2400 steps. Through controlled experiments and an ablation study, I demonstrated that:

Width is the key factor which increasing width improves performance; reverting width restores the baseline score exactly.

Depth and head count do not help under the fixed training budget.

A boundary configuration (W=512 + 2400 steps) achieved even lower validation BPB (1.606) but was rejected because its evaluation time exceeded the 5 $\times$ limit.

The final submitted model is width=256, heads=4, depth=4, 2400 steps, with validation BPB = 1.689 and test BPB = 1.7117.

AI Assistance Disclosure

In accordance with the assignment policy, I acknowledge the use of AI assistance (ChatGPT/DeepSeek) for environment setup guidance, experiment design suggestions, and report drafting. All experiments, code modifications, data analysis, and conclusions were independently performed and verified by me.